

Exponential, Logarithmic, and Inverse Functions

Section 3R

I. Review of Inverse Functions

A. Identifying One-to-One Functions

A function $f(x)$ is one-to-one if every element in the range corresponds to only one element in the domain.

If $f(a) = f(b)$ then $a = b$ or if $a \neq b$ then $f(a) \neq f(b)$

Horizontal Line Test: If there is NO horizontal line that intersects the graph more than once, then the function is one-to-one.

Example: Determine whether each function is one-to-one.

1. $f(x) = x^2 + 2$

2. $g(x) = \frac{x}{x-2}$

B. Inverse Functions

Let $f(x)$ be a function that is one-to-one and that goes through the point (a, b)

- Then $f^{-1}(x)$ is the inverse of $f(x)$
- $(f \circ f^{-1})(x) = x$
- $f^{-1}(x)$ will go through the point (b, a)
- The domain of $f(x)$ = the range of $f^{-1}(x)$
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C. Finding Inverse Functions

Steps: 1. Test to see whether the function is one-to-one

2. Replace $f(x)$ with y

3. Interchange x and y

4. Solve equation for y

5. Replace y with $f^{-1}(x)$

Example: Verify that the functions are inverse of each other

1. $f(x) = \sqrt[3]{x+4}$ and $g(x) = x^3 - 4$

Example: Find the inverse for each of the following:

1. $k(x) = (x-1)^3$

2. $g(x) = \frac{2x-3}{x+1}$

3. $h(x) = x^2 + 3$

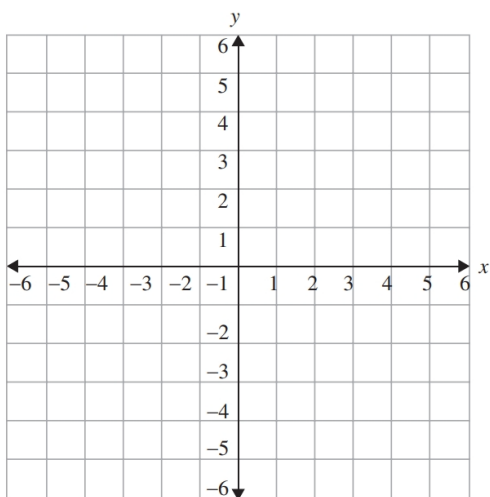
D. Graphs of Inverse Functions

The graph of $f^{-1}(x)$ can be constructed by mirroring the graph of $f(x)$ over the line $y = x$

Examples

- Construct the graph of $f^{-1}(x)$ if $f(x) = \sqrt{x}$
- The following are points on the graph of $f(x)$: $(2, 10)$, $(-1, -3)$, $(0, -2)$, $(1, -1)$, $(3, 6)$

Find at least five points on the graph of $f^{-1}(x)$

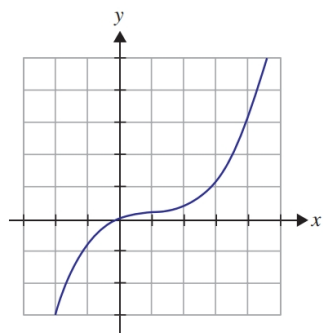


D. Domain and Range of Inverse Functions

The domain of $f(x)$ = the range of $f^{-1}(x)$

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Example: A function $f(x)$ has the following graph. Find the domain and range of the inverse function $f^{-1}(x)$.



II. Review of Exponential and Logarithmic Functions

A. Exponential Functions

DEFINITION: An exponential function is a function in the form $f(x) = a^x$. (i.e., the variable x is in the exponent)

Example: Find x for each of the following:

1. $4 = 2^x$

2. $27 = 3^x$

3. $2 = 4^x$

B. Logarithmic Functions

I. Logarithmic Functions

A logarithm is a function that helps us to solve a quadratic function/logarithms allow us to isolate the variable in a quadratic function (and the other way around).

DEFINITION: A logarithmic function is a function in the form $f(x) = \log_a x$ (i.e., the variable x is in the expression) $y = \log_b x$ “ y is equal to log base b of x ” - Here “ b ” is the BASE NUMBER and “ x ” is the VARIABLE.

$$\log_b x = y \text{ means exactly the same thing as } b^y = x$$

Examples: Write each equation in its equivalent form:

1. $x = \log_2 16$

2. $y = \log_6 216$

3. $3 = \log_b 27$

4. $8^y = 300$

II. Common Logarithmic Properties

1. $\log_b b = 1$

2. $\log_b 1 = 0$

3. $\log_b 0 = DNE$

4. $\log_b b^x = x$

5. $b^{\log_b x} = x$

6. $\log x = \log_{10} x$

Example: Simplify Each Expression

1. $\log_2 2 =$

2. $\log_6 1 =$

3. $\log_4 4^x =$

4. $\log_z z^{(x+y)} =$

5. $8^{\log_8(12y)} =$

6. $\log 10 =$

*III. The Natural Logarithm*DEFINITION: e is a number that equals approximately 2.718281828

$\log_e x = \ln x$

Example

1. $\log_e(3z) =$

2. $\ln e =$

IV. Expansion Properties for Logarithms

1. $\log_b(M \cdot N) = \log_b M + \log_b N$ (Product Rule)

2. $\log_b\left(\frac{M}{N}\right) = \log_b M - \log_b N$ (Quotient Rule)

3. $\log_b(M^n) = n \cdot \log_b M$ (Power rule)